

DeciMark: A Johnny.Decimal Bookmark Manager

Lyra Phasma | ITEC50 Final Project

An Individual Submission for the "Full-Stack Concepts" MVP

Phase 1: Research & Requirements

The Void in Bookmark Management

- **Pinboard:** Minimalist, flat tags, visually dead.
- **Raindrop.io:** Visually rich, heavily nested, lacks structural discipline.
- **DeciMark's Solution:** Enforcing the Johnny.Decimal numeric classification system (Area > Category > ID) natively.

Phase 2: Design & The Cascade (CSS)

Algorithmic Layouts & The Box Model

- **CSS Grid:** Used `repeat(auto-fill, minmax(280px, 1fr))` for a fluid, breakpoint-free dashboard layout.
- **Flexbox:** Used for single-axis alignment (navigation, card metadata).
- **Box Model:** Global `box-sizing: border-box` to prevent padding/margins from breaking flex alignments.

Phase 2: Specificity & Inheritance

Overcoming Cascade Conflicts

- **The Challenge:** Styling the SVG icon system globally, while allowing contextual overrides.
- **The Solution:** A base `.svg-icon` class (low specificity), overridden by a contextual selector `.bookmark-card .icon-delete` to safely change the `background-color` without resorting to `!important`.

Phase 3: Media Optimization & Typography

Vector Exclusivity & Zero-Trust Fonts

- **SVG Engine:** Zero raster images. Icons are injected via the CSS `mask-image` property.
- **HSL Theming:** Light/Dark mode is achieved by adjusting *only* the Lightness variable in CSS custom properties.
- **Local Typography:** Built a custom Python pipeline to compile TrueType fonts into ultra-compressed WOFF2 binaries. Zero reliance on Google Fonts.

Phase 4: System Logic (JavaScript)

Data Handling & DOM Output

- **Reference Types:** Client-side state is a module-scoped Array of Bookmark objects.
- **Conditional Logic & Loops:** `Array.prototype.filter()` evaluates the active state, feeding a `for...of` loop to render the UI.
- **DOM Output:** Raw `document.createElement()` dynamically builds cards, strictly bypassing `innerHTML` to prevent XSS.

Reflection: Client-Server & The Cascade

- **Client-Server Reality:** Built a *Pessimistic UI*. Deleting a bookmark doesn't immediately remove it—it dispatches an HTTP request. The UI only updates when the server confirms a 204 No Content status.
- **Cascade Conflict:** Managing CSS variable updates for dark mode. Solved by injecting the theme state directly into the `<html>` element via Jinja2 before the first paint to prevent CSS flashing.

PART II: BACKEND ARCHITECTURE

Moving beyond the rubric into Enterprise Engineering

Zero-Trust Cryptography

An Impenetrable Authentication Layer

- **Argon2id:** Passwords subjected to memory-hard key derivation to resist GPU brute-forcing.
- **Double-Encrypted Sessions:** JWTs are symmetrically encrypted with **Fernet** before being injected into `HttpOnly`, `SameSite=Lax` cookies.
- **Data at Rest:** Sensitive columns (URLs, tags) are encrypted using SQLAlchemy `TypeDecorator`.

High-Performance Data Persistence

Bypassing ORM Bottlenecks

- **FastAPI + SQLAlchemy:** Eliminating dual class definitions; a single class serves as both DB schema and API DTO.
- **asyncpg Engine:** Asynchronous PostgreSQL driver for sub-millisecond query execution.

High-Performance Data Persistence

Bypassing ORM Bottlenecks

- **Eager Loading:** Rigorous use of `selectinload` to completely eradicate N+1 queries.
- **Absolute Normalization:** Strict junction tables and cascading foreign keys. Orphan records are mathematically impossible.

Infrastructure & Engineering Discipline

Production-Grade Tooling

- **Deployment:** Containerized via Docker Compose behind a Netbird Reverse Proxy.
- **Schema Management:** Alembic handles all deterministic database migrations.

Infrastructure & Engineering Discipline

Production-Grade Tooling

- **Strictness:** Ruff and `stylelint` cranked to maximum constraints for zero-defect velocity.
- **Testing:** Playwright E2E visual testing orchestrated with hardware-aware concurrency limits.

Contingency: The Offline Fallback

Because live demos always break

- **PyInstaller:** The entire application is compiled into a standalone executable.
- **SQLite Fallback:** Automatically detects the offline environment and falls back from PostgreSQL to a local `sqlite:///decimark.db`.
- **In-Memory Cache:** Bypasses Redis seamlessly for local rate-limiting and session management.

Conclusion & Live Demo

DeciMark proves that absolute structural discipline—from the frontend CSS Grid to the encrypted PostgreSQL backend—creates a fundamentally superior user experience

Thank you.